

Negative Exponents with Variable Bases

Teacher's Edition

The Period

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Must Integer exponents on monomial bases, negatives included: rewrite so that only positive exponents are left, and combine several laws in one expression.

Must not Variables are nonzero wherever a negative exponent appears. No rational exponents, and monomials only.

Target I can rewrite an algebraic expression with negative exponents using only positive exponents, and apply several laws of exponents to a monomial expression.

MTR MTR.5.1 — Notes I · **MTR.6.1** — Board 4 · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	13 min
Independent Set	33	6 min
IXL	34	5 min
Total		53 min

Say these three things out loud today

1. A negative exponent on a letter means what it meant on a number: reciprocal.
2. Only the factor wearing the negative exponent moves. In $3x^{-2}$, the 3 stays upstairs.
3. Simplified means no negative exponents are left. Check every one before the board goes up.

Slide by Slide

1. **Negative Exponents with Variable Bases.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q2 and Q4 together are the lesson: a negative exponent asks for a reciprocal.
3. **Warm-Up — Answers.** Reveal. 1. $27x^6$ 2. $1/25$ 3. 9 4. $\frac{3}{2}$
4. **Notes — Copy the table.** x^{-3} is one over x^3 . It is a reciprocal. It is not negative, and it is not $-3x$.
5. **Notes.** Only the factor wearing the negative exponent moves. Everything else stays put.
6. **Notes.** Product and quotient laws are unchanged; the exponents are integers now, so the sums and differences can land below zero.
7. **Example 1 — Multiplying monomials with negative exponents.** SPOKEN ONLY.
8. **Example 1 — Worked.** Add the exponents of each variable, then move the y^{-2} down.
9. **Your Turn — Same steps, your numbers. Boards up when done.** Catches the coefficient-mover ($a^3/(4b^2)$) and the sign-dropper ($4a^7b^4$).
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — Several laws in one expression — the state guide's item.** SPOKEN ONLY.
12. **Example 2 — Worked.** Raise every factor to the 5th: a^{15} on top, $4^5 = 1,024$ and b^{-35} on the bottom.
13. **Your Turn — Same steps, your numbers. Boards up when done.** Catches the un-raised coefficient ($x^6y^6/3$) and the b-that-did-not-move ($x^6/(27y^6)$).
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** $x^{-8} \rightarrow \frac{1}{x^8}$ $-x^8$ — negative exponent read as a negative expression [B1G-M 8.AR.1.1 misconception: negative exponents]; $-8x$ — exponent multiplied into the base
- 17-18. **Board 2.** $\frac{1}{y^{-9}} \rightarrow y^9$ $1/y^9$ — the sign was changed but the power was not moved [8.AR.1.1: law applied in one direction only]
- 19-20. **Board 3.** The brightness of a lamp from x meters away is $5x^{-2}$ units. The lamp is 15 centimeters tall. How bright is the lamp from 3 meters away? $\rightarrow \frac{5}{9}$ The height of 15 centimeters is not needed. $1/45$ — the coefficient was moved down with the variable [8.AR.1.1: a negative exponent applies only to its own base]; 45 — the minus on the exponent was dropped, $5 \cdot 3^2$; -45 — negative exponent read as a negative value [B1G-M 8.NSO.1.3: $-b$ vs b^{-1}]; anything built from 15 — the unneeded figure was used
- 21-22. **Board 4.** $p^{-7}q^{-2} \rightarrow \frac{1}{p^7q^2}$ $1/(pq)^9$ — exponents on different bases were added [8.AR.1.1: like bases only]; $-p^7q^2$ — read as a negative
- 23-24. **Board 5.** $x^3 \cdot x^{-14}$ Which expression is equivalent? $\rightarrow \mathbf{C - 1/x^{11}}$ (A) x^{11} read as the opposite of x^{11} [B1G-M 8.AR.1.1 misconception: negative exponents]; (B) $3 \cdot$

(-14) — multiplied the exponents [power of a power confused with a product]; (D) 3 - (-14) — subtracted instead of adding a negative [product law: add the exponents] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

25-26. Board 6. $(3x^{-5})^2$ Which expression is equivalent? → **D — $9/x^{10}$** (A) $-5 + 2 = -3$ instead of $-5 \cdot 2$ [power of a power confused with a product of powers]; (B) the coefficient was moved below the bar with the x [8.AR.1.1: a negative exponent applies only to its own base]; (C) the coefficient was not squared [8.AR.1.1: power of a product] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

27-28. Board 7. $\frac{a^2}{a^{11}}$ → $\frac{1}{a^9}$ $a^9 - 11 - 2$ instead of $2 - 11$ [quotient law: top minus bottom]; $a^9 -$ correct but not in the positive-exponent form asked for

29-30. Board 8. Find the value of n : $x^{-5} \cdot x^n = x^2$ → **$n = 7$** -3 — added the given exponents [8.AR.1.1: product law, solve $-5 + n = 2$]; -7 — sign error in the integer subtraction

31-32. Board 9. How are x^5 and x^{-5} related? Explain, and give the value of each when $x = 2$. → **x^{-5} is the reciprocal of x^5 . When $x = 2$: $x^5 = 32$ and $x^{-5} = 1/32$.** ' $x^{-5} = -32$ ' or 'they are opposites' — negative exponent read as a negative [B1G-M 8.AR.1.1 misconception: negative exponents]; ' -10 ' — exponent multiplied into the base

33. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

34. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. $-x^8$ — negative exponent read as a negative expression [B1G-M 8.AR.1.1 misconception: negative exponents]

Board 2. $1/y^9$ — the sign was changed but the power was not moved [8.AR.1.1: law applied in one direction only]

Board 3. $1/45$ — the coefficient was moved down with the variable [8.AR.1.1: a negative exponent applies only to its own base]

Board 4. $1/(pq)^9$ — exponents on different bases were added [8.AR.1.1: like bases only]

Board 5. x^{-11} read as the opposite of x^{11} [B1G-M 8.AR.1.1 misconception: negative exponents]

Board 6. $-5 + 2 = -3$ instead of $-5 \cdot 2$ [power of a power confused with a product of powers]

Board 7. $a^9 - 11 - 2$ instead of $2 - 11$ [quotient law: top minus bottom]

Board 8. -3 — added the given exponents [8.AR.1.1: product law, solve $-5 + n = 2$]

Board 9. ' $x^{-5} = -32$ ' or 'they are opposites' — negative exponent read as a negative [B1G-M 8.AR.1.1 misconception: negative exponents]

Materials Whiteboards and markers. Calculators are allowed on everything; $4^5 = 1,024$ in Example 2 is the one computation worth a calculator.